

ST ANDREW'S JUNIOR COLLEGE



JC2 Preliminary Examinations

Chemistry

Higher 1

8872/01

Paper 1 Multiple Choice

19 September 2012

50 minutes

Additional Materials: Multiple Choice Answer Sheet, Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil. Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **30** questions on this paper. Answer all questions. For each question there are four possible answers A, B, C and D. Choose the one you consider correct and record your choice in soft pencil on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **14** printed pages **including** this page.

[Turn Over

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct.

1. Which of the following contains 1 mole of the stated particle?

- A** Nitrogen molecules in 14 g of nitrogen gas.
- B** Oxygen atoms in 24 dm³ of oxygen gas at room temperature and pressure.
- C** Hydrogen ions in 1000 cm³ of 1 mol dm⁻³ aqueous sulfuric acid.
- D** Electrons in 1 g of hydrogen gas.

2. Vanadium metal can be used to remove an excess of chlorine from bleaching processes by reducing it to chloride ions.

0.102 g of vanadium required 96 cm³ of chlorine for complete reaction at room temperature and pressure. In this reaction, what is the final oxidation number of the vanadium-containing product?

- A** +2
- B** +3
- C** +4
- D** +5

3. Which of the following particles has a half-filled p subshell after losing two electrons?

- A** Ga⁻
- B** As²⁺
- C** Se⁻
- D** Te⁺

4. The second ionisation energy of oxygen is 3390 kJ mol^{-1} .
Which of the following correctly represents this information?
- A $\text{O}^+ (\text{g}) \rightarrow \text{O}^{2+} (\text{g}) + \text{e}^-$
B $\text{O} (\text{g}) \rightarrow \text{O}^{2+} (\text{g}) + 2 \text{e}^-$
C $\text{O}^- (\text{g}) + \text{e}^- \rightarrow \text{O}^{2-} (\text{g})$
D $\text{O} (\text{g}) + 2 \text{e}^- \rightarrow \text{O}^{2-} (\text{g})$
5. Which of the following pairs of compounds has the same molecular shape?
- A CO_2 and SO_2
B BCl_3 and BrCl_3
C BeF_2 and BrF_2^-
D ClF_4^- and PF_4^-
6. Element X is in Group V of the Periodic Table. It forms salts which contain the XF_4^{n-} anion, the shape of which is square planar. What is the value of n?
- A 2
B 3
C 4
D 5
7. Which of the following has the highest enthalpy change of vapourisation?
- A $\text{CH}_3\text{COOCH}_2\text{CH}_3$
B $\text{CH}_2(\text{OH})\text{CH}_2\text{CH}_2\text{COOH}$
C $\text{CH}_3\text{CH}_2\text{COONH}_4$
D $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_2\text{CH}_2\text{NH}_2$
8. The enthalpy change of neutralisation of aqueous sodium hydroxide by aqueous

hydrochloric acid is $-57.2 \text{ kJ mol}^{-1}$, but the enthalpy change of neutralisation of aqueous sodium hydroxide by aqueous ethanoic acid is $-55.2 \text{ kJ mol}^{-1}$.

Which statement best explains the numerical difference in the above values?

- A** Aqueous ethanoic acid contains fewer hydrogen ions than aqueous hydrochloric acid of the same concentration.
- B** The neutralisation between aqueous sodium hydroxide and aqueous ethanoic acid produces less amount of water.
- C** The dissociation of aqueous ethanoic acid is endothermic.
- D** A larger volume of aqueous ethanoic acid is required for the neutralisation compared to that of aqueous hydrochloric acid of the same concentration.

- 9.** In an experiment to determine the enthalpy change of neutralisation, 20 cm^3 of 1 mol dm^{-3} of aqueous sulfuric acid was placed in a polystyrene cup of negligible heat capacity.

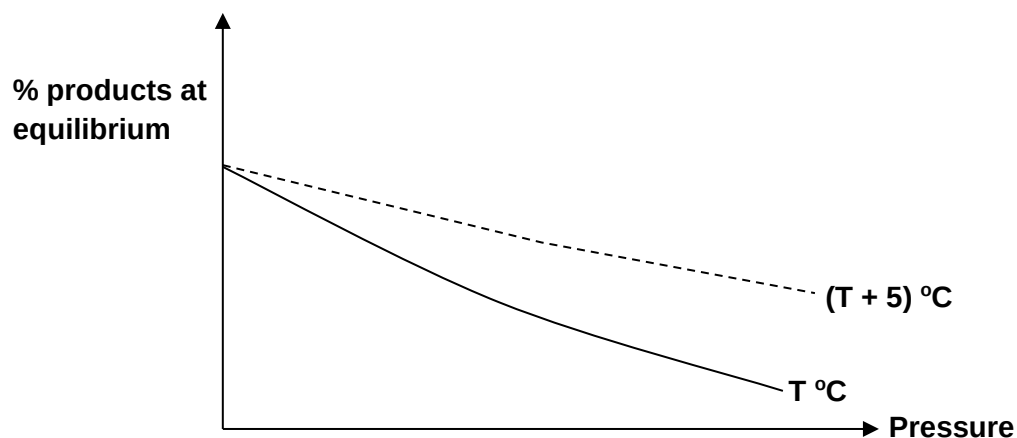
20 cm^3 of 2 mol dm^{-3} of aqueous potassium hydroxide, at the same initial temperature, was added and the temperature rose by 15°C .

If the specific heat capacity of the final solution is $4.2 \text{ J K}^{-1} \text{ cm}^{-3}$, what is the enthalpy change of neutralisation?

- A** $-\left(\frac{20 \times 4.2 \times 15}{0.04 \times 1000}\right) \text{ kJ mol}^{-1}$
- B** $-\left(\frac{20 \times 4.2 \times 15}{0.02 \times 1000}\right) \text{ kJ mol}^{-1}$
- C** $-\left(\frac{40 \times 4.2 \times 15}{0.04 \times 1000}\right) \text{ kJ mol}^{-1}$
- D** $-\left(\frac{40 \times 4.2 \times 15}{0.02 \times 1000}\right) \text{ kJ mol}^{-1}$

- 10.** The graph below shows the variation of the percentage of gaseous products present at

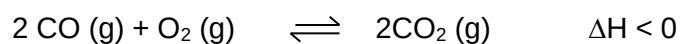
equilibrium with pressure and two temperatures, $T\text{ }^{\circ}\text{C}$ and $(T + 5)\text{ }^{\circ}\text{C}$.



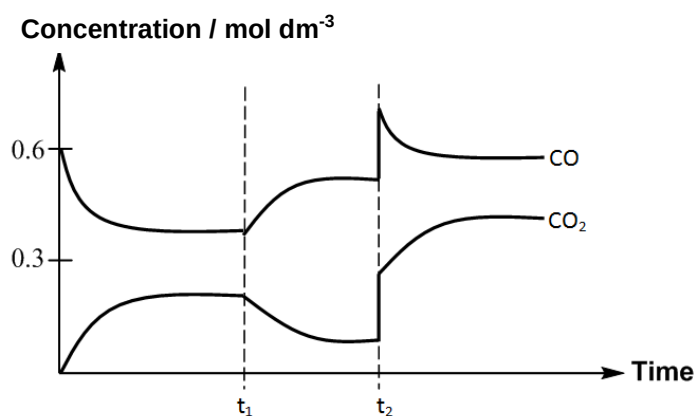
Which one of the following systems could the graphs represent?

- | | | |
|----------|---|---------------------------------------|
| A | $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ | $\Delta H = -92\text{ kJ mol}^{-1}$ |
| B | $3\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g}) \rightleftharpoons 2\text{N}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ | $\Delta H = -1248\text{ kJ mol}^{-1}$ |
| C | $2\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{N}_2\text{O}(\text{g})$ | $\Delta H = +82\text{ kJ mol}^{-1}$ |
| D | $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g})$ | $\Delta H = +173\text{ kJ mol}^{-1}$ |

- 11.** At a temperature $T\text{ K}$, 0.60 mol dm^{-3} of CO and 0.30 mol dm^{-3} of O_2 were introduced into a 5 dm^3 vessel and allowed to reach equilibrium.



The graph below shows the changes in the concentration of CO and CO₂ in the system with time. A change was made to the system at time, t_1 and t_2 .



What were the changes made at time, t_1 and t_2 ?

	t_1	t_2
A	A catalyst was added	Volume of the system is increased
B	The temperature was increased	Volume of the system is decreased
C	More CO ₂ was added	An inert gas was added at constant volume
D	The temperature was decreased	More O ₂ was added

12. Solutions **P**, **Q**, **R** and **S** contain a strong monobasic acid, a weak monobasic acid, a strong monoacidic base and a weak monoacidic base, but not necessarily in the same order. The concentration and pH of each solution are shown below.

Solution	Concentration /mol dm ⁻³	pH
P	0.1	4.0
Q	0.1	9.4
R	0.1	13.0
S	0.1	1.0

Which one of the following statements is **false**?

- A** **P** contains a weak acid while **S** contains a strong acid.
- B** Mixing 10 cm³ of **P** and 5 cm³ of **R** produces a buffer solution.
- C** Mixing 10 cm³ of **P** and 10 cm³ of **S** produces a buffer solution.
- D** Mixing 10 cm³ of **Q** and 10 cm³ of **S** produces an acidic solution.

13. How much water must be added to 5.0 dm³ of a solution of a strong acid of pH 2.0 in order to increase the pH to 3.0?

- A** 5.0 dm³
- B** 30.0 dm³
- C** 45.0 dm³
- D** 50.0 dm³

14. Which statement describes the effect of a catalyst on a reversible reaction?

- A** It increases the equilibrium constant for the forward reaction.
- B** It increases the rate constant for both the forward and reverse reactions.
- C** It increases the rate constant for the forward reaction but not that of the reverse reaction.
- D** It decreases the enthalpy change of the forward reaction.

15. The rate equation for the reaction $2\text{T}(\text{aq}) + \text{U}(\text{aq}) \rightarrow \text{V}(\text{aq})$ is

$$\text{rate} = k[\text{H}^+][\text{T}]^2$$

Which of the following statements **cannot** be deduced from the information above?

- A When the concentration of **T** is tripled, the rate increases by 6 times.
- B The units for the rate constant is $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$.
- C When the concentration of **U** is halved, the rate remains unchanged.
- D H^+ acts as a catalyst in the reaction.

16. An experiment was carried out to determine the rate equation for the following reaction.



The experimental results are tabulated below.

Experiment	Volume of L /cm ³	Volume of M /cm ³	Volume of water /cm ³	Relative initial rate of reaction
1	10	20	20	1
2	20	20	10	2
3	10	30	10	1
4	30	60	10	<i>y</i>

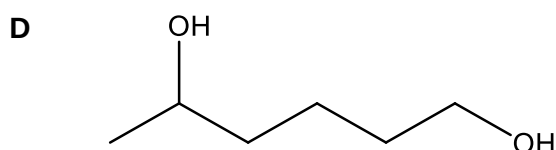
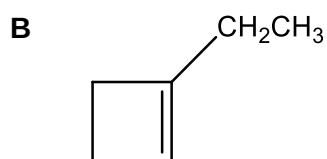
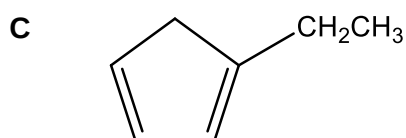
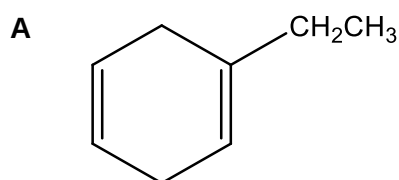
What is the value of *y*?

- A 1
 - B 1.5
 - C 3
 - D 9
17. The oxide and chloride of an element **F** are separately mixed with water. The two resulting solutions have the same effect on methyl orange and can also be used to dissolve BaO. What is element **F**?
- A Magnesium
 - B Aluminium
 - C Silicon
 - D Phosphorus
18. Which of the following statements regarding elements in Period 3 and their compounds is **incorrect** as atomic number increases?
- A The electronegativity of elements gradually decreases.

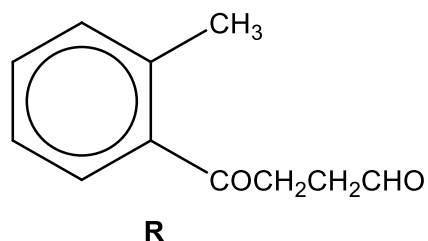
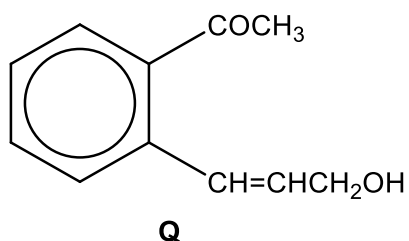
- B** The atomic radius of the elements decreases.
C The solutions of their oxides become increasingly acidic.
D The solutions of their chlorides become increasingly acidic.

19. An organic compound, on heating with an excess of hot concentrated acidified $\text{KMnO}_4(\text{aq})$, produces $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_2\text{COOH}$ as one of the products.

Which of the following shows the structure of the organic compound?



20. Compounds **Q** and **R** are isomers.



Which of the following reagents will distinguish **Q** from **R**?

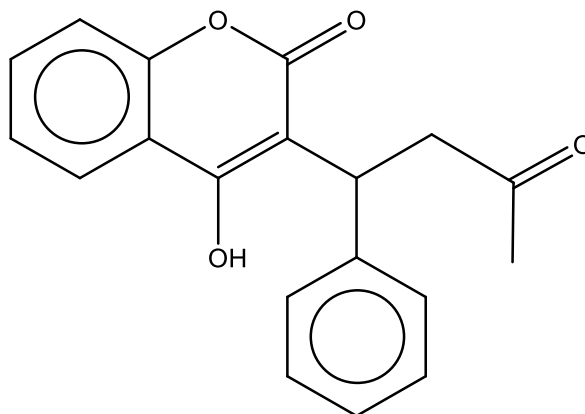
- | | |
|---------------------------------------|--|
| A hot acidified dichromate(VI) | C LiAlH_4 in dry ether |
| B Fehling's solution | D $\text{HCl}(\text{g})$ |

21. How many cyclic hydrocarbons are there with the molecular formula C_4H_6 ?

- A** 2
B 3

- C 4
D 5

22. Warfarin is used as a pesticide against rodents. The structural formula of warfarin is given below.

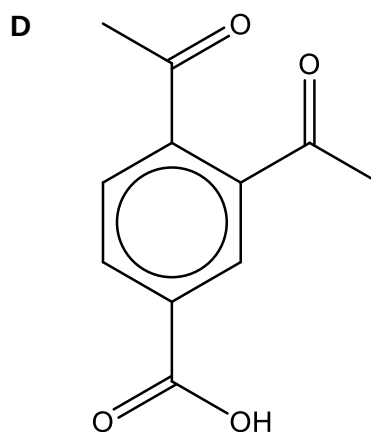
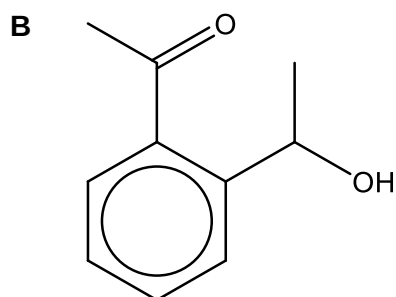
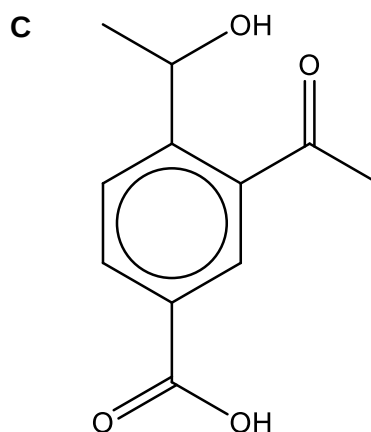
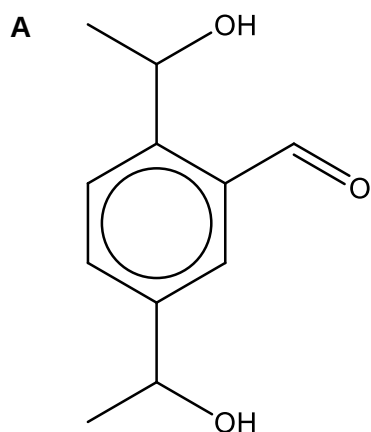


Which of the following reactants will **not** react with warfarin?

- A Aqueous bromine
B Warm aqueous alkaline iodine
C 2,4-DNPH
D Hot ethanolic KCN
23. Which of the following shows the compounds arranged in a trend of decreasing K_a ?
- A $\text{CH}_2\text{FCH}_2\text{COOH}$, $\text{CH}_3\text{CHFCH}_2\text{COOH}$, $\text{CH}_3\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CHFCH}_2\text{OH}$
B $\text{CH}_3\text{CHFCH}_2\text{COOH}$, $\text{CH}_2\text{FCH}_2\text{COOH}$, $\text{CH}_3\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CHFCH}_2\text{OH}$
C $\text{CH}_3\text{CHFCH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{COOH}$, $\text{CH}_2\text{FCH}_2\text{COOH}$, $\text{CH}_3\text{CHFCH}_2\text{COOH}$
D $\text{CH}_3\text{CH}_2\text{COOH}$, $\text{CH}_3\text{CHFCH}_2\text{OH}$, $\text{CH}_2\text{FCH}_2\text{COOH}$, $\text{CH}_3\text{CHFCH}_2\text{COOH}$
24. Compound **J** exhibits the following characteristics:
- 1 mol of **J** reacts with excess sodium metal to give 1 mol of H_2 .
 - 1 mol of **J** reacts with aqueous alkaline iodine to give 2 mol of CHI_3 .
 - Orange precipitate is formed when **J** reacts with 2,4-DNPH, but no reaction with

alkaline silver diammine solution.

What could compound **J** be?



- 25.** Which of the following alcohols will give the most number of products when it is heated with excess concentrated H_2SO_4 ?

- A** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
B $\text{C}(\text{CH}_3)_3\text{OH}$
C $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
D $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$

Section B

[Turn Over

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

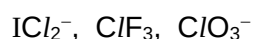
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26. For the elements **X** and **Y** with atomic number 14 and 15 respectively, which of the following statements are true?

- 1** The fifth ionisation energy of **X** is higher than that of **Y**.
- 2** They can form XCl_4 and YCl_3 respectively.
- 3** The atomic radius of **X** is smaller than that of **Y**.

27. For the following chlorine-containing species,



Which of the following statements are **incorrect**?

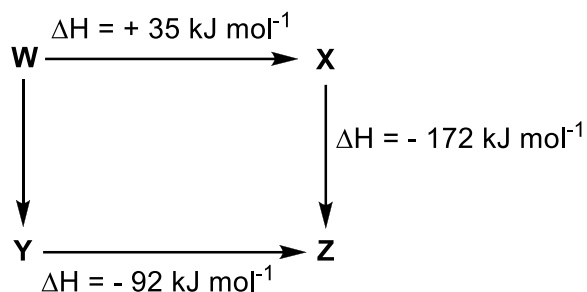
- 1** All atoms achieve stable octet configuration.
- 2** ICl_2^- is linear, ClF_3 is T-shaped while ClO_3^- is trigonal pyramidal.
- 3** The bond angles increase in the order of ClF_3 , ClO_3^- , ICl_2^-

A	B	C	D
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1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct
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No other combination of statements is used as a correct response.

28. The diagram below illustrates the energy changes for a set of reactions.



Which of the following statements are correct?

- 1 The enthalpy change for the transformation $W \rightarrow Z$ is -137 kJ mol^{-1} .
 - 2 The transformation $Y \rightarrow X$ is endothermic.
 - 3 The enthalpy change for the transformation $W \rightarrow Y$ is -45 kJ mol^{-1} .
29. Methyl propanoate $\text{CH}_3\text{OCOCH}_2\text{CH}_3$ undergoes acidic hydrolysis in the presence of H_2^{18}O . Which of the following products are formed?
- 1 CH_3OH
 - 2 $\text{CH}_3\text{CH}_2\text{CO}^{18}\text{OH}$
 - 3 $\text{CH}_3\text{CH}_2^{18}\text{OH}$

A	B	C	D
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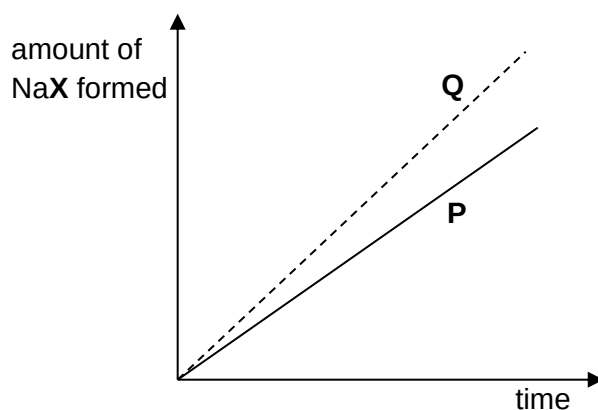
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1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct
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No other combination of statements is used as a correct response.

30. When a halogen-containing compound is hydrolysed with aqueous NaOH, the corresponding sodium halide, NaX, is produced.

A student investigated the number of moles of NaX produced by hydrolysing two halogen-containing compounds, **P** and **Q**, separately. The experimental results obtained are shown graphically below.



Which compounds could be **P** and **Q**?

	P	Q
1	$\text{CH}_3\text{CH}_2\text{Cl}$	$\text{CH}_3\text{CH}_2\text{Br}$
2	$\text{C}_6\text{H}_5\text{CH}_2\text{Br}$	$\text{BrCH}_2\text{CH}_2\text{Br}$
3	$\text{CH}_3\text{CH}_2\text{Br}$	$\text{CH}_2=\text{CHBr}$

~~~ END ~~~